

ai ex-11.py - C:/Users/chaith/AppData/Local/Programs/Python/Python313/ai ex-11.py (3.13.5)

File Edit Format Run Options Window Help

```
from itertools import permutations

letters = ('S', 'E', 'N', 'D', 'M', 'O', 'R', 'Y')
digits = range(10)

for perm in permutations(digits, len(letters)):
    S, E, N, D, M, O, R, Y = perm

    # Leading letters cannot be zero
    if S == 0 or M == 0:
        continue

    SEND = 1000*S + 100*E + 10*N + D
    MORE = 1000*M + 100*O + 10*R + E
    MONEY = 10000*M + 1000*O + 100*N + 10*E + Y

    if SEND + MORE == MONEY:
        print("Solution Found!")
        print("S =", S)
        print("E =", E)
        print("N =", N)
        print("D =", D)
        print("M =", M)
        print("O =", O)
        print("R =", R)
        print("Y =", Y)

        print("\nSEND =", SEND)
        print("MORE =", MORE)
        print("MONEY =", MONEY)
        break
```

IDLE Shell 3.13.5

File Edit Shell Debug Options Window Help

Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>> ===== RESTART: C:/Users/chaith/AppData/Local/Programs/Python/Python313/ai ex-11.py =====

Solution Found!

S = 9

E = 5

N = 6

D = 7

M = 1

O = 0

R = 8

Y = 2

SEND = 9567

MORE = 1085

MONEY = 10652

>>>

Ln: 18 Col: 0

Ln: 31 Col: 13